



Characterization and Transferability of *erm* and *tet* Antibiotic Resistance Genes in *Lactobacillus* spp. Isolated from Traditional Fermented Milk

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Abstract

Lactobacillus is a widely used bacteria and consumed through various fermented foods and beverages. Strains have been shown to carry resistance genes and mobile genetic elements with their ability to transfer the resistance to sensitive pathogenic strains. To study this, 4 cultures of *Lactobacillus* were isolated from traditional fermented milk. The isolates were able to grow up to 4% (w/v) NaCl concentration and 45 °C temperature, and showed > 97% 16S rRNA gene similarities with *Lactobacillus fermentum*. All the isolates were phenotypically screened for the presence of antibiotic resistance. Minimum inhibitory concentration (MIC) as microbiological breakpoints were observed against a varied class of antibiotics. Isolates AKO 94.6, DVM 95.7, and NIFTEM 95.8 were explicitly resistant to ampicillin, ciprofloxacin and vancomycin with MIC well beyond the maximum range of 256 µg/ml in the E-strip test. While isolate SKL1 was sensitive to ampicillin and showed MIC at 0.25 µg/ml but resistant to streptomycin and trimethoprim (MIC > 256 µg/ml). Molecular characterization showed the presence of *tet*(M) gene in three isolates SKL1, DVM 95.7, and NIFTEM 95.8 which was chromosomally associated resistance determinants while *erm*(B) resistance gene was detected in isolates DVM 95.7 and NIFTEM 95.8 only which was a plasmid associated gene and could be transferrable conjugally. Gene for Tn916 family (*xis*) was also observed in isolates DVM 95.7 and NIFTEM 95.8. Transferability of antibiotic resistance to pathogenic recipient strains was examined in isolates DVM 95.7 and NIFTEM 95.8 in different food matrices. The highest conjugation frequency with $\sim 10^{-1}$ was obtained in alfalfa seed sprouts. This study reports the presence of acquired gene resistance in *Lactobacillus* species and dissemination to susceptible strains of bacteria in different food matrices. 16S rRNA gene sequences of isolates were uploaded to the NCBI GenBank database to retrieve the accession number.

Introduction

Fermented foods and beverages have been consumed since ancient times across the world. Lactic acid bacteria (LAB) comprising the genera *Lactobacillus*, *Leuconostoc*,

Lactococcus, *Enterococcus* and *Pediococcus* play a significant role in food fermentation. These organisms can be found in both humans' and animals' gastro-intestinal tract (GIT) as a common inhabitant. Species of lactobacilli are regarded as non-pathogenic and have been given GRAS (Generally Recognized As Safe) status owing to their enormous applications in fermented foods for centuries [1]. These are currently used as probiotic organisms in commercial fermented products as well as pharmaceuticals. Probiotics are defined as “live microorganisms which when administered in adequate amounts, confer health benefits to the host” [2]. Probiotics impart diverse fitness advantages to consumers consisting of lactose intolerance relief, avoidance of many types of diarrhea and urogenital infections, cholesterol depletion, atopic disease reduction, and immune system modulation [3–8]. Other advantages include cancer prevention, particularly colon carcinoma and food allergy avoidance. Recently their function as

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antioxidants has additionally been reported [7]. Role of probiotics in the alleviation of diabetes have been delineated by Madempudi and their co-workers recently in the year 2022 [9]. Various species of *Lactobacillus* e.g. *L. acidophilus*, *L. casei*, *L. rhamnosus*, *L. helveticus* are considered successful probiotics.

The selection of suitable probiotic organisms is done through a range of in vitro and in vivo medico-scientific and technological parameters such as acid and bile tolerance, adherence to the intestine, and storage stability during low-temperature storage. The organism must also be safe in terms of being non-toxic and must not harbor transferable antibiotic resistance (AR) genes [10]. The bacteria may contain AR genes and can potentially transfer the genes to other lactic acid bacteria as well as pathogenic bacteria [11–13]. According to European Food Safety Authority (EFSA), bacterial strains which have the ability to transfer the AR genes should be discouraged to use in animal feeds and fermented probiotic foods [14]. The emergence of AR is of significance all over the world. The consequence of antibiotic resistance includes increased morbidity and mortality, and prolonged hospitalization, associated with an increased cost in health care. The role LAB can play in AR and the transfer of resistance genes down the food chain has not been studied in depth. Some strains of *Lactobacillus* spp. are resistant to vancomycin, bacitracin, metronidazole, norfloxacin, streptomycin, and clindamycin [15]. Horizontal gene transfer amongst LAB, via self-transmissible plasmids and conjugative transposons, is a common phenomenon, yet not well studied [16]. The plasmid pLFE1 showing resistance to erythromycin from *L. plantarum* strain M345, was successfully transferred to *L. rhamnosus*, *Lactococcus lactis*, *Listeria innocua*, *Enterococcus faecalis*, and *Listeria monocytogenes* by filter-mating experiments [17]. The acquired AR genes on plasmid pK 214 in *Lactococcus lactis* shared high homology to AR genes present in the pathogens *S. aureus* and *Listeria monocytogenes* [18]. Some strains of *Lactobacillus* spp. resistant to tetracycline and erythromycin acquired the genes (*tetM*, *tetS*, and *ermB*) from LAB present in meat and other fermented foods [19, 20]. This shows that AR is transferrable between pathogens in food and LAB that may inhabit the human GIT.

In the present study, 4 different isolates of *Lactobacillus fermentum* (SKL1, DVM 95.7, NIFTEM 95.8, and AKO 94.6), out of a pool of 40 presumptive isolates were characterized. These cultures were then investigated for the existence of tetracycline and erythromycin resistance gene determinants using E-strip and PCR method to find out the genetic basis of resistance. The ability to transfer these antibiotic resistance was further assessed by in vitro filter mating technique and also by in situ transfer in different food matrices.

Materials and Methods

Isolation, Identification and Preservation

Four bacterial cultures were isolated from fermented milk (Curd, equivalent to yoghurt) obtained from local households. Isolation was accomplished using the standard plating technique. Suitable dilutions were plated on MRS (de Man, Rogosa and Sharpe, Hi-Media Ltd., Mumbai, India) agar and typical colonies were selected. Preliminary identification was carried out by microscopic and biochemical examinations. Isolates were subjected to both the catalase and oxidase test. For growth in anaerobic conditions, an Anaero Gas Pack jar (Hi-media) was used. Gram-positive rods and catalase/oxidase-negative isolates were taken for further identification based on carbohydrate fermentation (KB009 HiCarbo Kit, Hi-Media). Isolates were preserved in 25% (v/v) glycerol stock medium at -80°C . All four isolates were designated as SKL1, DVM 95.7, NIFTEM 95.8, and AKO 94.6. The recipient organisms used for the conjugation experiment were *E. faecalis* ATCC 14506, *E. coli* ATCC 11229 and *S. aureus* NCDC 109. All recipients and transconjugants were grown on Brain Heart Infusion (BHI) agar media (Table 3).

Physiological Parameter Study

Optimum growth temperature and salinity conditions were studied using MRS broth. Temperature ranges were selected from 20 to 50°C while salinity ranges were from 0 to 10% (w/v). For temperature optima, cultures were grown in shake flasks by keeping them at 20 , 25 , 30 , 35 , 40 , 45 , and 50°C for 48 h at 150 rpm in a shaker incubator (Kuhner—ISF1-X, Switzerland) under aerobic conditions. Whereas optimum NaCl concentration was determined by growing the cultures at 37°C for 48 h in an aerobic environment. To study the optimum pH, biological buffers from pH 3.0 to 9.0 (with an interval of 1.0) were used. For pH 3.0 to 7.0 , 0.1 M citric acid/ 0.2 M Na_2HPO_4 were used as buffer medium for culture medium preparation while for pH 8.0 , 0.2 M Na_2HPO_4 / 0.2 M $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ were used and for pH 9.0 , 0.1 M Na_2CO_3 / 0.1 M NaHCO_3 were used [21]. All cultures were incubated at 37°C for 48 h aerobically.

Isolation of Genomic and Plasmid DNA

Isolation of complete genomic DNA was carried out manually as per the protocol followed earlier with slight modifications [22]. Briefly, overnight grown bacterial culture (1.5 ml) was centrifuged at $6010 \times g$ (Eppendorf 5424R, Germany). The pellet was washed in $1\times$ PBS (phosphate buffer saline,

pH 7.0) and centrifuged again at 6010 $\times g$. In the microcentrifuge tube (2 ml), 0.5 ml of SET buffer (75 mM NaCl, 25 mM EDTA, 20 mM Tris, pH 7.5) and 100 μ l of lysozyme (1 mg/ml) were added to the cell pellet and incubated for an hour at 37 °C. Following incubation, 50 μ l of 10% (w/v) SDS was added, followed by 10 μ l of proteinase K, and incubated for 2 h at 56 °C with intermittent inversion. To dissolve the proteins and separate cell debris, 550 μ l chloroform and 180 μ l of 5 M NaCl were added to the tube and inverted repeatedly at regular intervals for 30 min at room temperature. Centrifugation of the tube was done at 9391 $\times g$ for 15 min. To precipitate the DNA, the uppermost aqueous phase was taken into a new tube and isopropanol (chilled) was added. The tube was centrifuged for 7 min at 9391 $\times g$. The pellet was subjected to washing (Five times) with 70% (v/v) ethanol and subsequently dried to remove any residual ethanol. TE buffer (100 μ l) was used to dissolve DNA and incubated at 37 °C overnight. DNA was stored at -20 °C for long time preservation. Plasmid DNA was isolated by GeneJET Plasmid Miniprep Kit (Thermo Scientific, Waltham, Massachusetts, USA), as per the instructions given by the manufacturer.

All four isolates were further subjected to molecular characterization by genus-based identification of isolates for confirmation as members of lactobacilli as per the procedures mentioned by Dubernet and his co-workers in 2002 [23]. The DNA was PCR amplified in a thermal cycler using genus-specific primers LbLMA1 and R16-1 (Table 1). The programmed conditions of initial denaturation at 95 °C for 5 min was followed by 35 cycles of denaturation at 95 °C for 1 min, annealing at 55 °C for 1 min, amplification at 72 °C

for 1 min and a final extension at 72 °C for 8 min (Eppendorf, Germany). The amplified DNA product was resolved on a 0.8% (w/v) agarose gel stained with ethidium bromide (0.5 μ g/ml) through gel electrophoresis at 80 V for 30 min using 1X TAE buffer (Mini/ Maxi submarine, Hoeffer, USA). Gels were imaged after being monitored using a UV Transilluminator (Gel capture, DNR Bio-Imaging System). The isolates with a $\approx$ 250 bp amplicon size were confirmed as lactobacilli genus.

16S rRNA gene amplification was carried out by amplifying the genomic DNA using specific gene primer sets (Table 1) in thermal PCR with the program specified as earlier [24]. The amplified PCR products were purified and sequenced through outsourcing from Chromous Biotech (Bengaluru, India). The raw reads of 16S rRNA gene sequences were analyzed using NCBI GenBank database BLAST search and phylogenetic dendrogram reconstruction was done using MEGA version 7.0 to check the placements of our strains in a phylogenetic tree [25, 26].

Determination of Susceptibility Pattern

The MIC of selected isolates was evaluated using antibiotic E—strips (Hi-Media Ltd., Mumbai, India) against 14 clinically important antibiotics i.e. ampicillin, chloramphenicol, ciprofloxacin, clindamycin, erythromycin, gentamycin, kanamycin, linezolid, rifampicin, streptomycin, tetracycline, tigecycline, trimethoprim, and vancomycin. The concentrations in all the antibiotics were ranging from 0.016 to 256 μ g/ml except rifampicin and trimethoprim where concentration was used from 0.002 to 32 μ g/ml.

Table 1 Primers used during PCR amplification

Target gene region	Primer sequence (5'-3')	Product size (bp)	Annealing temp (°C)	References
LbLMA1	F: CTCAAACTAAACAAAGTTTC	~250	55	[23]
R16-1	R: CTTGTACACACCGCCCGTCA			
16S rRNA	63f: CAG GCC TAA CAC ATG CAA GTC 1387r: GGG CGG WGT GTA CAA GGC	~1300	55	[24]
<i>erm</i> (A)	F: AAGCGGTAAACCCCTCTGAG R: TCAAAGCCTGTCGGAATTGG	~441	62	[52]
<i>erm</i> (B)	F: CATTTAACGACGAAACTGGC R: GGAACATCTGTGGTATGGCG	~425	61	[52]
Tet(M)	F: ACACGCCAGGACATATGGAT R: CTTTCCGCAAAGTTCAGAC	~537	60	[52]
Tet(O)	F: ACACGCCAGGACATATGGAT R: TCCCACTGTCCATATCGTCA	~519	60	[52]
Tet(S)	F: GGAGTACAGTCACAAACTCG R: GGATATAAGGAGCAACTTTG	~335	53	[52]
<i>Int</i>	F: GCGTGATTGTATCTCACT R: GACGCTCCTGTTGCTTCT	~1028	50	[45]
<i>xis</i>	F: AAGCAGACTGAGATTCCTA R: GCGTCCAATGTATCTATAA	~194	50	[45]

100 µl of overnight grown cultures of all organisms containing cell density as 10^8 CFU/ml (0.5 Mcfarland) was taken and using a sterile swab spread over MRS agar plate uniformly. The antibiotic strips were put at the center of the agar plate surface and it was ensured that the strip was uniformly in contact with the agar surface. MIC values were noted after 24 h of incubation of plates at 37 °C. The organisms were classified as resistant or susceptible based upon EFSA guidelines as described earlier [27–30].

Detection of Antibiotic Resistance and Transposon Genes

The AR genes for the determination in this study were: tet(M), tet(O), tet(S) erm(A) and erm(B). While for detection of transposon genes, isolates were examined for conjugative transposon Tn916 family genes (*Int* and *xis* gene). These genes were detected by PCR amplification using specific gene primer sequences (Table 1). Reaction for PCR amplification was set up with a volume of 25 µl using a 2X DreamTaq PCR master mix (Thermo Scientific, Waltham, Massachusetts, USA), 10 pmol of primers, and template DNA. Amplification of reaction was done with the following program: initial denaturation at 95 °C for 4 min, followed by 40 cycles of denaturation at 95 °C for 30 s, annealing temperature from 55 to 65 °C (as per Tm of individual primers) for 30 s, amplification at 72 °C for 1.5 min and a final extension at 72 °C for 7 min. PCR products amplified were resolved by electrophoresis on 1.2% (w/v) agarose gel and observed under gel documentation system (Gel capture, DNR Bio-Imaging System).

In Vitro Filter Mating Transfer of Resistance Genes

Conjugal transfer of resistance genes was performed in vitro through filter mating techniques using the procedures as illustrated earlier [31]. Briefly, from an actively grown overnight culture of the donor (*Lactobacillus* strains) and recipient, a fresh media broth was inoculated with 2% (v/v) of the cultures (adjusted to 0.5 Mcfarland turbidity standard) in tubes containing respective media for donor and recipient culture and incubated for 4–5 h. After a brief incubation, in a microcentrifuge tube of 1.5 ml, donor and recipient cultures were taken in a ratio of 1:10 and centrifuged at 6010 ×g for 10 min at 4 °C. The collected pellet was suspended in 1 ml of sterile peptone physiological saline (PPS) solution and the resulting suspension was allowed to filter through a 0.45 µ pore-sized sterile membrane filter (Nupore Filtration System Pvt. Ltd., India). The membrane was incubated at 37 °C for 24 h on Brain Heart Infusion agar (BHI) media. From the membrane, cells were suspended in 1 ml peptone physiological saline solution (PPS) with the help of a vortex mixer and the suspension was serially diluted. Appropriate dilutions

were spread plated on selective media for donor/ recipient, and transconjugants and incubated at 37 °C for 24–72 h.

Conjugal Mating in food Matrices (In Situ Mating)

Conjugal Transfer In Fermented Milk

Conjugal mating in the food matrix was carried out using the methods described earlier with slight modifications [20, 32]. To assess the conjugal mating in fermented milk, skim milk (10% w/v) was dissolved in 100 ml distilled water and autoclaved (121 °C for 5 min, 15 psi) before inoculation. In case of fermented milk, 1 ml (10^8 CFU/ml) of donor and recipient each were mixed in equal ratio and then centrifuged at 6010×g for 10 min at 4 °C to pellet down the cells. These cell pellets were suspended in 1 ml of skim milk and the whole suspension was inoculated further in 100 ml of sterile fresh skim milk to allow for milk fermentation at 37 °C until the pH dropped to around 4.5–4.0. After the milk was coagulated, it was chilled and kept at 4 °C for 2 days. A serial dilution of fermented milk was made and plated according to the procedures followed as in the above filter mating experiments (“[In vitro filter mating transfer of resistance genes](#)” section).

Mating in Fermented idli Batter

Idli is a fermented food consumed popularly as a savory rice cake in the Indian subcontinent especially in Southern regions of India. To prepare idli batter for fermentation and conjugal mating, the protocol adopted by Thumu and Halami [20] was followed with some modifications [20]. 10 ml of the batter was inoculated with 1 ml (10^8 CFU/ml) of each respective donor and recipient culture and incubated at 37 °C for 24 h. The inoculated batter was refrigerated at 4 °C after 24 h of incubation to stop over fermentation for a week. Diurnally, aliquots of each sample were serially diluted and suitable dilutions were plated on a selective medium with appropriate antibiotics to evaluate transconjugants (TCs) for a period of seven days.

Mating in Fermented Minced Chicken

To determine the in situ conjugative gene transfer between *Lactobacillus* species and food pathogens, the fresh chicken was homogenized (Polytron® PT 3100D, Kinematica AG) and minced into the batter with certain modifications in protocols as followed by researchers previously [20, 33]. The two donor strains, *L. fermentum* DVM 95.7 and *L. fermentum* NIFTEM 95.8 were used. The recipient strains used are mentioned in Table 3. Briefly, the minced chicken batter was kept in UV radiation for 30 min to get rid of the extraneous bacteria existing in the raw meat. The viable bacterial

cells were also checked for their absence by plating them on MRS as well as BHI agar media plates. A sterile solution consisting of 3% NaCl (w/v), 1% glucose (w/v) and 0.017% of nitrite (w/v) was added further to the batter of chicken and mixed thoroughly. This resulting batter of meat was distributed equally into a 100 ml sterile sample collecting container (Hi-Media). Each container was inoculated with a 1:1 ratio of the donor and recipient strains (10^8 CFU/gm) and fermented for a day at 23 °C, followed by incubation for 27 days at 15 °C. During the fermentation process, meat was sampled on days 0, 2, 7, 14, 21 and 28. Physiological saline (PPS) was used to make a serial dilution of each sample. TCs were determined by plating dilutions on BHI agar medium containing selective antibiotics (Table 3).

Conjugative Transfer in Plant Model

Alfalfa seeds were used as a plant model for transferability studies [32, 34]. 4 g of alfalfa sprouting seeds were taken in a sterile container aseptically. 3 ml (10^8 CFU/ml) of each donor and recipient culture strains were used to inoculate alfalfa seeds. A control consisting of only the seeds was set up by soaking it with 6 ml of autoclaved distilled water. Seeds were then kept at an average temperature of 22 °C to get soaked overnight and incubated further for 9 days with intermittent sampling to enumerate transconjugants on selective culture media. Deionized sterile water (2 ml) was used to maintain the moisture level in the sprouts at regular intervals. At the interval of 0, 1, 3, 6, and 9 days, 10 sprouts or seeds were taken in a sterile stomacher bag followed by the addition of 3 ml PPS and pulsification for 1 min. The suitable dilutions were then plated onto selective agar media plates using the spread plate method to enumerate transconjugants. All plates were incubated at 37 °C for 48 h aerobically.

Results and Discussion

Four bacterial cultures isolated from different curd samples (AKO 94.6, DVM 95.7, NIFTEM 95.8, and SKL1) were taken up for further studies. The isolates were identified as the species of genus *Lactobacillus* through morphological, biochemical, and molecular (16S rRNA & genus-specific) characterization. A microscopic study showed that the isolates AKO 94.6, DVM 95.7, NIFTEM 95.8, and SKL1 were Gram-positive bacilli arranged singly, in pair, or chains under 100X objective of the microscope (Fig. 1). The other physiological, biochemical, and antibiotic susceptibility tests are given in Table 2. All these four isolates were subjected to phenotypic AR detection and MIC determination using the E-strip method. The isolates showed variable responses against different antibiotics it was found that isolates AKO

94.6, DVM 95.7, and NIFTEM 95.8 were resistant to ampicillin and ciprofloxacin while the isolate SKL1 was found sensitive to ampicillin but resistant to ciprofloxacin antibiotic. Resistance to erythromycin was observed in the isolate AKO 94.6 and NIFTEM 95.8 while the other two isolates were susceptible to erythromycin (DVM 95.7 and SKL1). Phenotypic resistance was also observed against tetracycline in isolates AKO 94.6 and SKL1 but no resistance was detected in the isolates DVM 95.7 and NIFTEM 95.8 against tetracycline. Isolates AKO 94.6, DVM 95.7 and NIFTEM 95.8 showed resistance to erythromycin. A pattern of multi-drug resistance was observed in our study where these isolates were resistant to more than five antibiotic drugs. The MICs breakpoint (cut-off) values of all four isolates against other classes of antibiotics were noted and the sensitivity/resistance profile was evaluated according to the guidelines provided by EFSA (2012) and Danielsen & groups in 2003 as shown in table (Table 2).

Genomic characterization through genus-specific PCR gene amplification establishes that isolates AKO 94.6, DVM 95.7, NIFTEM 95.8, and SKL1 fell under the genus *Lactobacillus*, which was confirmed by doing a BLAST with 16S rRNA gene sequences of these isolates in the NCBI GenBank database. BLAST results showed that these isolates showed a match with *Lactobacillus fermentum* with a similarity between 97 and 99% proving that all four strains belonged to the species of *L. fermentum*. Phylogenetic dendrogram of 16S rRNA gene sequences of isolates showed that all four isolates (AKO 94.6, DVM 95.7, NIFTEM 95.8, and SKL1) fell in the same clade of the tree with *L. fermentum* species. A high bootstrap similarity percentage value confirms the closed similarity among the isolates with *L. fermentum* (Fig. 2). The sequences of the 16S rRNA gene of all four isolates were also submitted in NCBI GenBank. The accession numbers were retrieved as MW647658(AKO 94.6), MW647600(DVM 95.7), MW647659(NIFTEM 95.8) and MW647661(SK1).

Isolates DVM 95.7, NIFTEM 95.8, and SKL1 showed tet(M) resistance gene in PCR amplification reaction while isolates DVM 95.7 and NIFTEM 95.8 showed the presence of erm(B) gene (Fig. 3a, b). The other isolate AKO 94.6 did not show any positive amplification for the presence of these two genes but it was observed to show resistance against both erythromycin and tetracycline phenotypically. The tetracycline gene tet(M) detected was chromosomally associated while erythromycin gene erm(B) was found to be present in plasmid DNA of *Lactobacillus* strains. Due to the probable link of tet(M) gene to conjugative transposon Tn916, it could be transferred to susceptible species of *Lactobacillus* or other pathogenic bacteria present in the GI (gastro intestinal) tract [35–40]. We observed the presence of *xis* (excisionase) gene of Tn916 class of transposon in our study in isolates DVM 95.7 and NIFTEM 95.8 only (Fig. 3c).

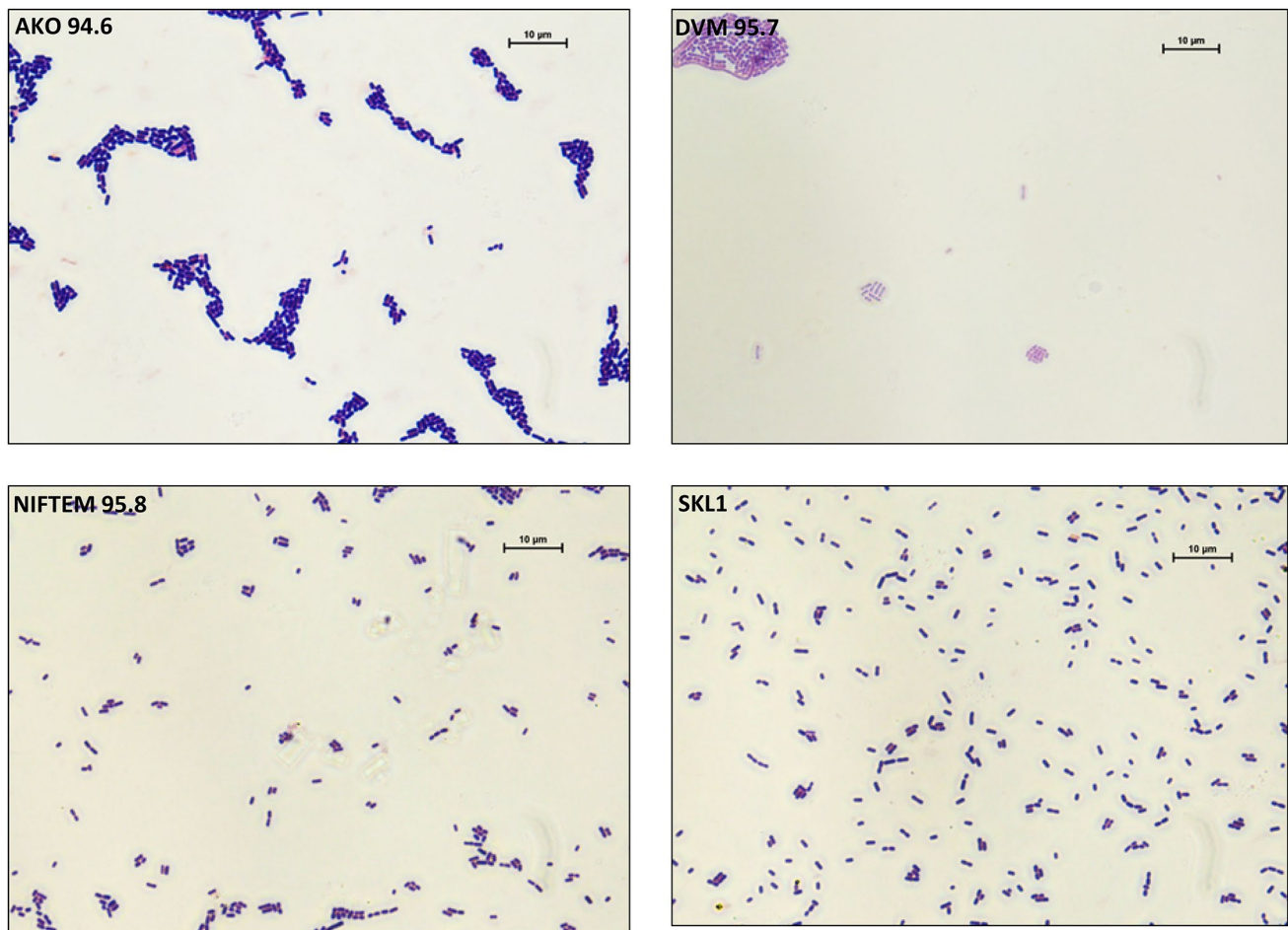


Fig. 1 Gram's staining images of the isolates under 100X objective

Here in our study interestingly it is discernible that isolate DVM 95.7 did not show resistance to erythromycin phenotypically but harbored the *erm(B)* gene which could be transferred conjugatively to the susceptible strains of lactobacilli as well as to food pathogens. The presence of *erm(B)* gene was one of the most frequently occurring genes of erythromycin resistance determinant in *Lactobacillus* species [36, 38, 41, 42]. Whereas *tet(M)* gene was reported recently by some group of researchers in *Lactobacillus* species [38, 43]. The resistance gene determinant *erm(B)* codes for the enzyme rRNA methylase which acts on the 23S ribosomal subunit while gene *tet(M)* codes for ribosomal protection proteins [38]. It was reiterated by several other researchers about the presence of ribosomal protection proteins coding genes and macrolide resistance in different *Lactobacillus* strains [20, 44–49]. Isolate NIFTEM 95.8 showed phenotypic resistance to erythromycin as well as the presence of the *erm(B)* gene. The other resistance genes like *tet(S)*, *tet(O)*, *erm(A)* and *Int* genes were not detected in our study.

To check the transferability of the resistance gene determinants, in vitro and in situ methods were used. The filter

mating technique was employed for in vitro transfer of AR traits in selected mating pairs. The average transfer frequencies were determined and are given in Table 4. A total of two *Lactobacillus* isolates (DVM 95.7 and NIFTEM 95.8) containing a *tet(M)* and *erm(B)* gene were exploited to test their ability to transfer AR to the recipients *E. faecalis* ATCC 14506, *E. coli* ATCC 11229 and *S. aureus* NCDC 109 by conjugation. Many promising attempts were made to obtain transconjugants using a 0.45 µ filter membrane through the filter mating approach. The transfer frequency was 6.4×10^{-6} in the case of the mating pair of DVM 95.6 and *E. coli* ATCC 11229. Whereas, NIFTEM 95.8 when used as a donor with recipient *E. coli* ATCC 11229 gave a transfer frequency of 6.0×10^{-6} . The rate of transfer frequency is quite low here when compared to the frequency of transfer as reported in other recipient groups (10^{-3} TCs/recipients) [20, 50]. No transconjugants were observed onto the selective media plates when *E. faecalis* ATCC 14506 was used as a recipient. Transfer frequency was discerned in the recipient strain *E. faecalis* JH2-2 used by other researchers in the past [20, 31, 51].

Table 2 Characterization and detection of antibiotic resistance genes of strains AKO 94.6, DVM 95.7, NIFTEM 95.8 and SKL1

Characteristics	AKO 94.6	DVM 95.7	NIFTEM 95.8	SKL1
Isolation source	Curd	Curd	Curd	Indian <i>lassi</i>
Cell size (μm)	2.1×0.8	2.2×0.8	1.8×0.5	3.2×0.7
Oxidase	—	—	—	—
Catalase	—	—	—	—
Temperature range ($^{\circ}\text{C}$)	20–50	20–50	20–50	20–50
Temperature optimum ($^{\circ}\text{C}$)	35–40	35–40	35–40	35–40
Salinity range (%)	0–9	0–9	0–9	0–9
Salinity optimum (%)	0	0	0	0
pH range	3–9	3–9	3–9	3–9
pH optimum	6.5	6.5	6.5	6.5
HiCarbo Kit KB009				
Lactose	—	—	—	+
Xylose	—	—	—	—
Maltose	+	—	—	+
Fructose	—	—	—	+
Dextrose	—	—	—	+
Galactose	—	—	—	+
Raffinose	—	—	Weak positive	+
Trehalose	+	—	—	—
Melibiose	+	—	—	+
Sucrose	+	—	—	+
L-Arabinose	+	—	—	—
Mannose	+	—	—	—
Inulin	+	—	—	+
Sodium gluconate	—	—	—	—
Glycerol	—	—	—	—
Salicin	—	—	—	—
Dulcitol	—	—	—	—
Inositol	—	—	—	—
Sorbitol	—	—	—	—
Mannitol	—	—	—	—
Adonitol	—	—	—	—
Arabitol	—	—	—	+
Erythritol	—	—	—	—
α -Methyl-D-glucoside	—	—	—	—
Rhamnose	—	—	—	—
Cellobiose	—	—	—	—
Melezitose	—	—	—	Weak positive
α -Methyl-D-mannoside	—	—	—	—
Xylitol	—	—	—	—
ONPG	—	—	—	Weak positive
Esculin hydrolysis	—	Weak positive	—	—
D-Arabinose	—	—	—	—
Citrate utilization	—	—	—	—
Malonate utilization	—	—	—	—
Sorbose	—	—	—	—
MIC Breakpoints in $\mu\text{g/ml}$ (E-Strip)				
Ampicillin	R	R	R	S/0.25
Chloramphenicol	R/16	S/4	R/6	S/2
Ciprofloxacin	R	R	R	R
Clindamycin	S/0.094	S/0.125	S/0.094	S/0.19

Table 2 (continued)

Characteristics	AKO 94.6	DVM 95.7	NIFTEM 95.8	SKL1
Characteristics	AKO 94.6	DVM 95.7	NIFTEM 95.8	SKL1
Erythromycin	R/1.5	S/1	R/1.5	S/0.38
Gentamicin	S/8	S/12	S/12	S/4
Kanamycin	R/64	R	S/4	R
Linezolid	R/6	R/12	S/1.5	S/1.5
Rifampicin	S/0.032	S/0.094	S/0.094	S/0.125
Streptomycin	S/64	S/64	R/96	R
Tetracycline	R	S/8	S/8	R/12
Tigecycline	n.r/1	n.r/1	n.r/0.19	n.r/0.50
Trimethoprim	S/24	R	S/1.5	R
Vancomycin	n.r/8	R	R	R
Antibiotic resistance genes				
tet(M)	—	+	+	+
tet(S)	—	—	—	—
tet(O)	—	—	—	—
erm(A)	—	—	—	—
erm(B)	—	+	+	—
<i>Int</i>	—	—	—	—
<i>xis</i>	—	+	+	—

R resistant, *S* sensitive, *nr* not reported

Table 3 Bacterial strains and their selective media

Strains	Selective media	
	Donor/recipient	Transconjugants
Donors		
DVM 95.7 (<i>L. fermentum</i>)	MRS + 5 µg/ml Tet + 5 µg/ml Ery	
NIFTEM 95.8 (<i>L. fermentum</i>)	MRS + 5 µg/ml Tet + 5 µg/ml Ery	
Recipients		
<i>E. coli</i> ATCC 11229	BHI	BHI + 5 µg/ml Tet + 5 µg/ml Ery
<i>E. faecalis</i> ATCC 14506	BHI	BHI + 5 µg/ml Tet + 5 µg/ml Ery
<i>S. aureus</i> NCDC 109	BHI	BHI + 5 µg/ml Tet + 5 µg/ml Ery

In the food matrix, transferability was checked and transfer frequency was observed in minced chicken only with recipients *E. coli* ATCC 11229 and *E. faecalis* ATCC 14506 (Table 5). However, no transfer frequency was recorded in foods like fermented milk and *idli* batter. Here we achieved a transfer frequency of $\sim 10^{-2}$ TCs/recipients which seemed to be more than what was obtained in the filter mating experiment.

In the plant model (alfalfa), the transfer frequency obtained is given in Table 6. Transfer frequency was observed from day 6 onwards with two recipients only (*E. coli* ATCC 11229 and *E. faecalis* ATCC 14506). On day 9, no transfer frequency was determined except in strain

NIFTEM 95.8 which showed a frequency of 2.2×10^{-2} and 1.2×10^{-2} respectively with recipients *E. coli* ATCC 11229 and *E. faecalis* ATCC 14506. The highest AR transfer frequency obtained in our study was eventuated in the alfalfa sprout plant model (DVM 95.7, $\sim 10^{-1}$ TCs/recipient). This is indicating that this environment may be conducive for the optimal gene transfer in mating strains. Our results were in agreement with the mating experiment of the alfalfa plant model used in research work published by Toomey in 2009 [32]. The reason for getting such a high frequency in the plant might be the fact that the surface of the seed provides a stable environment which restricts the mobility of mating strains. This allows the mating pairs to get an intimate contact for conjugation to occur.

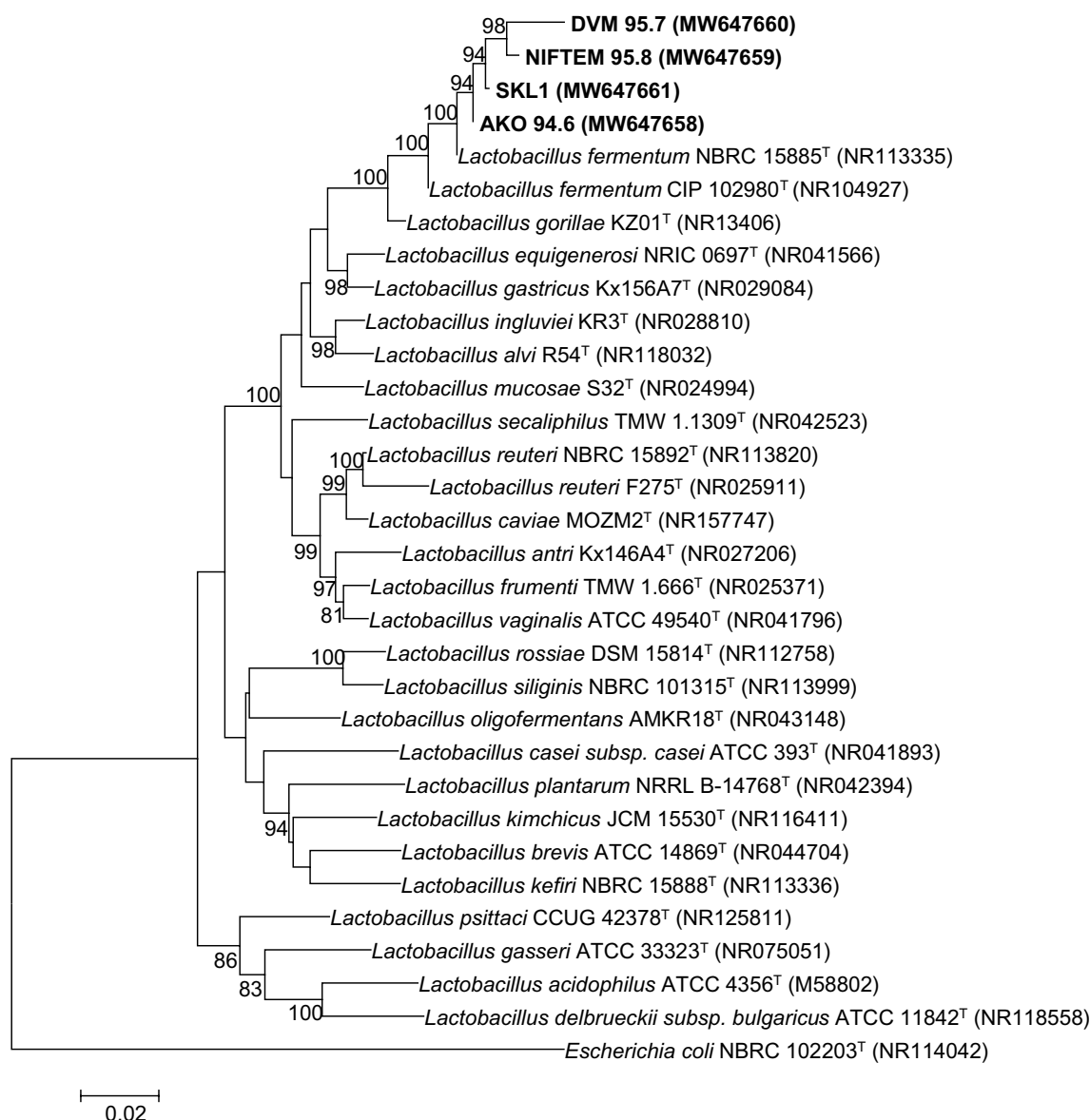


Fig. 2 Evolutionary relationships of taxa. The evolutionary history was inferred using the Neighbor-Joining method (Saitou and Nei 1987). The optimal tree with the sum of branch length=0.81070298 is shown. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) are shown next to the branches (Felsenstein, 1985). The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Maximum Composite

Likelihood method (Tamura et al. 2004) and are in the units of the number of base substitutions per site. The rate variation among sites was modeled with a gamma distribution (shape parameter=1). The analysis involved 32 nucleotide sequences. Codon positions included were 1st+2nd+3rd+Noncoding. All ambiguous positions were removed for each sequence pair. There were a total of 1541 positions in the final dataset. Evolutionary analyses were conducted in MEGA7 (Kumar et al., 2016). Only bootstrap values $\geq 70.0\%$ are shown

Conclusion

Our study shows the presence of transferrable AR genes in strains of *Lactobacillus* species. Here we have reported the highest number of transfer frequencies ($\sim 10^{-1}$) in plant model experiment of conjugation. Since tetracycline and erythromycin have been used in livestock feed at a

subtherapeutic level. Therefore, bacteria-harboring resistance genes might pose a risk of dissemination by horizontal gene transfer. Probiotics have several beneficial properties but due to their inadvertent use, they may lead to a potential risk to the health. This could lead to hazards as involved with the GI tract. Therefore, bacteria containing resistance genes as well as mobile genetic elements should not be preferred

Fig. 3 **a** PCR amplification of *erm* (B) gene. DVM 95.7 and NIFTEM 95.8. **b** PCR amplification of *tet* (M) gene. DVM 95.7; NIFTEM 95.8 and SKL1. **c** PCR amplification of *xis* gene. DVM 95.7 and NIFTEM 95.8

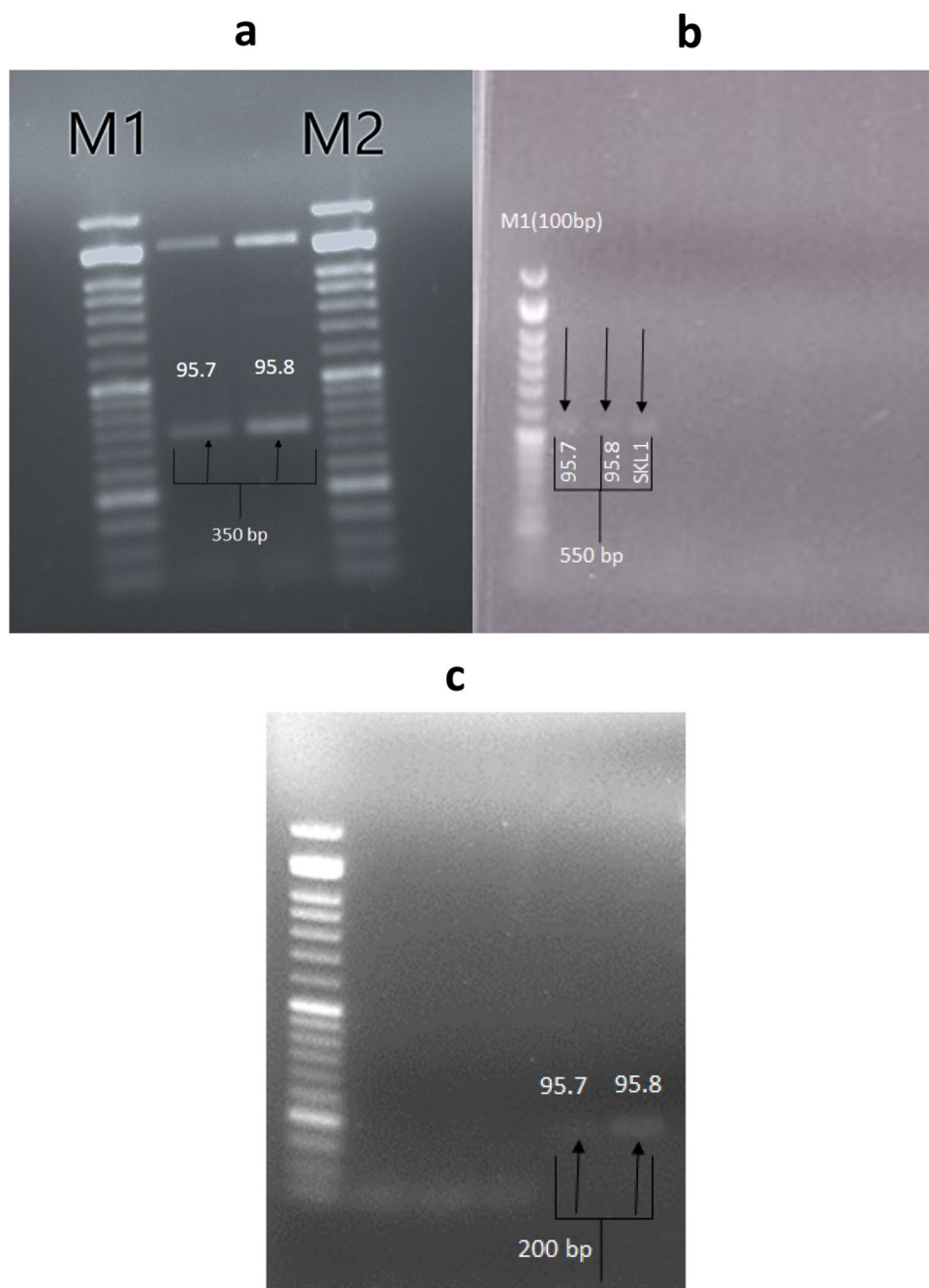


Table 4 Conjugative transfer through in vitro filter mating approach

Donor	Recipient	Genotype	Transfer frequencies (Number of TCs/No. of recipients)		
			In vitro	In situ (Fermented milk)	In situ (idli batter)
DVM 95.7 (<i>L. fermentum</i>)	<i>E. coli</i> ATCC 11229	Tet(M), <i>erm</i> (B)	6.4×10^{-6}	ND	ND
	<i>E. faecalis</i> ATCC 14506		ND	ND	ND
	<i>S. aureus</i> NCDC 109		ND	ND	ND
NIFTEM 95.8 (<i>L. fermentum</i>)	<i>E. coli</i> ATCC 11229	Tet(M), <i>erm</i> (B)	6.0×10^{-6}	ND	ND
	<i>E. faecalis</i> ATCC 14506		ND	ND	ND
	<i>S. aureus</i> NCDC 109		ND	ND	ND

Table 5 Conjugative transfer in minced chicken

Donor	Recipient	Genotype	Transfer frequency (no. of TCs/ no. of recipients)
DVM 95.7 (<i>L. fermentum</i>)	<i>E. coli</i> ATCC 11229	Tet(M), erm(B)	3.0×10^{-2}
	<i>E. faecalis</i> ATCC 14506		1.4×10^{-2}
	<i>S. aureus</i> NCDC 109		ND
NIFTEM 95.8 (<i>L. fermentum</i>)	<i>E. coli</i> ATCC 11229	Tet(M), erm(B)	2.0×10^{-2}
	<i>E. faecalis</i> ATCC 14506		1.0×10^{-2}
	<i>S. aureus</i> NCDC 109		ND

Table 6 Conjugative transfer in alfalfa plant model

Donor	Recipient	Genotype	Transfer frequency (no. of TCs/ no. of recipients)				
			Day 0	Day 1	Day 3	Day 6	Day 9
DVM 95.7 (<i>L. fermentum</i>)	<i>E. coli</i> ATCC 11229	Tet(M), erm(B)	ND	ND	ND	4.0×10^{-1}	ND
	<i>E. faecalis</i> ATCC 14506		ND	ND	ND	2.0×10^{-1}	ND
	<i>S. aureus</i> NCDC 109		ND	ND	ND	ND	ND
NIFTEM 95.8 (<i>L. fermentum</i>)	<i>E. coli</i> ATCC 11229	Tet(M), erm(B)	ND	ND	ND	3.2×10^{-2}	2.2×10^{-2}
	<i>E. faecalis</i> ATCC 14506		ND	ND	ND	1.8×10^{-2}	1.2×10^{-2}
	<i>S. aureus</i> NCDC 109		ND	ND	ND	ND	ND

to use as a starter in food processing. Hence to discourage these resistant bacteria such a study is imperative to proceed in this domain of food safety.

Author Contributions AKO contributed to conceiving, writing the manuscript, prepared figures. NPS contributed to refining the manuscript and VM conceived, revised the manuscript. All authors read and approved the manuscript.

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Declarations

Conflict of interest There is no declared conflict of interest.

Ethical Approval The paper does not contain any study on human participants or animals.

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